

Why Deep Learning? From Features to Representations

Scaler School of Technology | Deep Learning I | Week 1 — Session 1/1 | 29 April 2026



What Is This Lecture About?

This is your first deep learning lecture. We'll explore why classical machine learning hits a ceiling when dealing with complex data like images, audio, and text — and how deep learning breaks through that ceiling by learning its own features. You'll understand the theoretical foundation (Universal Approximation Theorem), see what neural networks actually learn inside, and build a framework for knowing when to use deep learning vs. classical ML.

1 Limitations of Hand-Crafted Features

Why feature engineering (SIFT, HOG, Bag-of-Words) works for tabular data but breaks down for images, audio, and text. The scaling problem: classical ML plateaus with more data.

2 Universal Approximation & Representation Learning

The theoretical guarantee that neural networks can learn any function. How deep networks build hierarchical features automatically — from edges to textures to full objects — without human intervention.

3 When DL Beats Classical ML & The DL Landscape

A practical framework: data size vs. data type. The ImageNet moment (2012) that changed everything. A preview of CNNs, RNNs, Transformers, and Generative Models — your roadmap for the next 9 weeks.

What to Expect in the Session

- Interactive discussion on feature engineering challenges
- Visual demos of what CNNs learn inside
- 2 short quizzes + a final 3-question quiz
- The AlexNet case study — how deep learning won ImageNet
- A hands-on Colab notebook with PyTorch exercises
- Course roadmap: what you'll learn in 18 sessions

How to Prepare (Optional but Helpful)

- Review what you know about feature engineering from Classical ML
- Recall how scikit-learn pipelines work (fit/predict)
- Think about: what makes image data different from tabular data?
- Optional: skim the Wikipedia article on the Universal Approximation Theorem
- Ensure you have Google Colab access with a GPU runtime
- No PyTorch experience needed — we'll start from basics

Duration: ~2 hours

Prerequisites: Classical ML, Advanced ML

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